

A – Multiple Choice · 1 mark each

1. Which one of the following is a **well-defined** set?

- (a) collection of beautiful girls in your class (b) the set of vowel letters of the English alphabet (c) collection of hardworking teachers in a school (d) collection of good basketball players

3. The set $A = \{1, 3, 5, 7, 9, \dots\}$ in set-builder form is

- (a) $\{x \mid x \in \mathbb{N}, x \text{ is even}\}$ (b) $\{x \mid x \in \mathbb{N}, x \text{ is odd}\}$ (c) $\{x \mid x \in \mathbb{N}, x > 9\}$ (d) $\{x \mid x \in \mathbb{Z}, x > 0\}$

5. The set of all even natural numbers is a

- (a) finite set (b) empty set (c) infinite set (d) universal set

7. $A = \{0, 1, 3, 5, 7\}$ and $B = \{1, 2, 3, 4, 6, 7\}$. Then $A \cup B =$

- (a) $\{0, 1, 3, 5, 7\}$ (b) $\{1, 3, 7\}$ (c) $\{0, 1, 2, 3, 4, 5, 6, 7\}$ (d) $\{2, 4, 6\}$

9. Given $U = \{1, 2, 3, 4, 5\}$ and $A = \{2, 4\}$, then $A' =$

- (a) $\{1, 3, 5\}$ (b) $\{2, 4\}$ (c) $\{1, 2, 3, 4, 5\}$ (d) $\emptyset$

11. Two sets A and B are **disjoint** if

- (a) $A = B$ (b) $A \subset B$ (c) $A \cap B = \emptyset$ (d) $A \cup B = \emptyset$

13. $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$. Then $A \setminus B =$

- (a) $\{3, 4\}$ (b) $\{1, 2\}$ (c) $\{5, 6\}$ (d) $\{1, 2, 3, 4, 5, 6\}$

15. $n(A) = 20$, $n(B) = 28$ and $n(A \cup B) = 36$. Then $n(A \cap B) =$

- (a) 48 (b) 12 (c) 8 (d) 20

17. In a Venn diagram, the region representing $A \cup B$ is shaded by

- (a) shading both circles completely (b) shading only the overlapping region (c) shading only one circle (d) shading the region outside both circles

19. If $n(A) = 5$, then the number of elements in the **power set** of A is

- (a) 10 (b) 25 (c) 32 (d) 16

B – Short Answer · 2 marks each

21. List **all** subsets of the set $\{a, b, c\}$.

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23. Give the set-builder form of $D = \{4, 8, 12, 16, \dots\}$.

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2. Given $A = \{2, 4, 6, 8, 10\}$, which statement is true?

- (a) $2 \notin A$ (b) $7 \in A$ (c) $4 \in A$ (d) $6 \notin A$

4. Which one of the following is an **empty set**?

- (a) $\{x \mid x \in \mathbb{N} \wedge 5 < x < 6\}$ (b) $\{0\}$ (c) $\{x \mid x \in \mathbb{N} \wedge x \leq 1\}$ (d) $\{2\}$

6. If $A = \{1, 2, 3\}$, the number of **proper subsets** of A is

- (a) 8 (b) 7 (c) 6 (d) 3

8. Using the sets in Q7, $A \cap B =$

- (a) $\{0, 2, 4, 6\}$ (b) $\{1, 3, 7\}$ (c) $\{0, 1, 2, 3, 4, 5, 6, 7\}$ (d) $\emptyset$

10. If $B = \{3, 4\}$ and $A = \{1, 2, 3, 4\}$, then

- (a) B is a proper subset of A (b) $A \subset B$ (c) $B \supset A$ (d) $A = B$

12. Sets A and B are **equal** if

- (a) $n(A) = n(B)$ (b) $A \subset B$ only (c) every element of A is in B and every element of B is in A (d) $A \cap B = \emptyset$

14. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A \nabla B =$

- (a) $\{3\}$ (b) $\{1, 2, 4, 5\}$ (c) $\{1, 2\}$ (d) $\{1, 2, 3, 4, 5\}$

16. The number of subsets of a set having 6 elements is

- (a) 12 (b) 32 (c) 64 (d) 128

18. If $X = \{2, 4, 6, 8\}$ and $Y = \{1, 4, 9\}$, then $n(X) + n(Y) =$

- (a) 7 (b) 8 (c) 9 (d) 16

20. For any set A , the complement of its complement, $(A')'$, is

- (a) A (b) A' (c) $\emptyset$ (d) U

22. Write $\{x \mid x \in \mathbb{N} \wedge 4 < x < 9\}$ using the complete listing (roster) method.

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24. $A = \{0, 2, 4, 6\}$ and $B = \{1, 2, 3, 5, 6\}$. Find (i) $A \cap B$ and (ii) $A \nabla B$.

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C – Problems · 3 marks each · show working

25. In a class of 60 students, 35 like football, 25 like basketball and 12 like both. (a) How many like exactly one sport? (b) How many like neither? (c) Sketch the Venn diagram.

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26. $A = \{2, 4, 6, 8, 10\}$, $B = \{1, 2, 3, 4, 5\}$. Find (a) $A \cup B$, (b) $A \cap B$, (c) $A \setminus B$, (d) $B \setminus A$, (e) verify $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

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27. Let $U = \{1, 2, 3, \dots, 12\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 3, 5, 7, 11\}$. Write down and shade: (a) A' , (b) $A \cap B$, (c) $A \cup B$, (d) $(A \cup B)'$.

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